Syllabus

Course: Operating Systems

Credits: 3

Course Designers:

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Aim of the course: This is a compulsory second year course, for all the branches of Bachelor of Technology that presents the elements of computer science engineering. The focus of this course is to teach the students to understand the concepts and features of Operating System.

Course Overview and Context: The course is meant to introduce students the fundamental concepts involved in operating system and how operating systems acts as an interface between the underlying hardware and the applications programs used by the user of a computing system. An understanding of what process is, how synchronization is done and how inter-process communication takes place. This course also covers memory, file and device management. Also, to deal with protection and security issues related to operating system.

Course Outcomes: At the end of the course the students should be able to	
CO1	Understand how application software interacts with hardware with the help of operating system
CO2	Implement various CPU scheduling algorithms
CO3	Analyze operating system with respect to process management, memory management etc
CO4	Evaluate the performance and features of current operating systems being used in industry and prepare a study document.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
Operating systems [Credits: 02]	*1*	*1*	*1*	*2*	*2*					*1*	*2*		*1*		*2*	

CO1	Understand how application software interacts with hardware with the help of operating system.	1	3			3								2		1	
CO2	Implement various CPU scheduling algorithms.	1	2	1		1								1		2	
CO3	Analyze operating system with respect to process management, memory management etc.		1			2								1		2	
CO4	Evaluate the performance and features of current operating systems being used in industry and prepare a study document.		1	2	2	2					1	2		1		2	

Course Competencies: (Course Outcomes further elaborated):

C1	Introduction to Operating system and types of operating system. Role of OS in real life applications	CO1

C2	Understand the concept of process, attributes related to process and operations performed on the process	CO2
C3	Discuss the process state diagram, role of context switching and dispatcher.	CO2
C4	Discuss why we need to perform CPU scheduling and under which condition this scheduling is done and goals related to it.	CO2
C5	Explain the different CPU Scheduling Algorithms: First Come First Serve, Shortest Job First, Round Robin Scheduling, Longest Job First, Highest Response Ratio Next, Priority Based Scheduling, Multilevel Queue Scheduling, and Multilevel Feedback Queue Scheduling.	CO2
C6	Understand how process synchronization is done and discuss what kind of problem arises not having synchronization between processes, conditions to be followed to achieve synchronization and solutions.	CO2
C7	Familiarize with the concept of concurrent programming	CO2
C8	Discuss characteristics, prevention, avoidance, detection and recovery of deadlock	CO2
C9	Explain thread, its advantages and categorization	CO2
C10	Discuss the contiguous and non-contiguous memory management techniques.	CO3
C 11	Familiarize with the concept of Virtual Memory and how it can be implemented.	CO3
C12	Discuss the page replacement algorithms: First In First Out, Optimal	CO3

	Page Replacement, Least Recently Used and Most Recently Used.	
C13	Discuss the role of file management, directory structure, operation	CO4
	and disk space allocation method	
C14	Explain the various disk scheduling algorithms.	CO4

Major Topics of the Course

Introduction to Operating Systems, Concept of batch-processing, multi-programming, time sharing, real time operations, Process Management : Scheduling, Threads, concept of semaphore, Process Synchronization, Deadlock; Memory management: partitioning, fragmentation, paging segmentation, swapping, virtual memory, demand paging, page size, page table, page replacement algorithms; File Management, disk ,

Learning Resources:

Text Books

- Operating Systems Concepts – A. Silberschatz, P. Galvin and G. Gagne. Wiley India

[Complete syllabus (100%) is being covered from the textbook]

Reference books (available in the library)

- Operating Systems Concepts - Gary Nutt, N. Chaki and S. Neogy, Pearson Education
- Operating Systems – W. Stallings, Pearson Education
- Operating Systems: A Concept-based Approach – D. M. Dhamdhere, Tata McGraw-Hill

Learning Online Resources

Will be shared when needed.

Assessment Pattern: The final grade will be based on the marks / grades obtained in the mid-semester and end-semester evaluation along with other assessments defined below. Relative grading method defined in the academic regulations of the university will be followed to grade the students. Student has to secure minimum 40% of marks after completing all the assessments in the following table to become eligible for grading.

Percentage Weightage	
Sessional	60 %
Finals	40 %

Sessional **Break up is as follows**:

Item	**Weightage (60%)**
Quiz 1	10%
Mid Sem	20%
Quiz 2	10%
Mini Project (working model)	20%

Attendance Policy: As per the University Policy.

Instruction schedule:

IU NO.	Competency	Scheduled No. of Lectures
IU1	Introduction to Operating system and types of operating system. Role of OS in real life applications	2
IU2	Understand the concept of process, attributes related to process and operations performed on the process	2
IU3	Discuss the process state diagram, role of context switching and dispatcher.	2
IU4	Discuss why we need to perform CPU scheduling and under which condition this scheduling is done and goals related to it.	2
IU5	Explain the different CPU Scheduling Algorithms: First Come First Serve, Shortest Job First, Round Robin Scheduling, Longest Job First, Highest Response Ratio Next, Priority Based Scheduling, Multilevel Queue Scheduling, and Multilevel Feedback Queue Scheduling.	4
IU6	Understand how process synchronization is done and discuss what kind of problem arises not having synchronization between processes, conditions to be followed to achieve synchronization and solutions.	4

IU7	Familiarize with the concept of concurrent programming	2
IU8	Discuss characteristics, prevention, avoidance, detection and recovery of deadlock	3
IU9	Explain thread, its advantages and categorization	1
IU10	Discuss the contiguous and non-contiguous memory management techniques	4
IU11	Familiarize with the concept of Virtual Memory and how it can be implemented.	1
IU12	Discuss the page replacement algorithms: First In First Out, Optimal Page Replacement, Least Recently Used and Most Recently Used.	1
IU13	Discuss the role of file management, directory structure, operation and disk space allocation method.	2
IU14	Explain the various disk scheduling algorithms.	2

Quiz / Test / Assignments: There will be multiple internal assessment components like Quiz – Surprise and Announced, Assignment – shared time to time and lab evaluation before the final exam

Evaluation procedures for tests and assignments: Practice problems will be shared via email as assignment. Each assignment will have a deadline for submission and each assignment will carry 5% of weightage. Every student should submit his/her assignment on or before the deadline given for that particular assignment.

Late assignment submission policy: Late submission in assignment is not allow ed and any late submission will be awarded "0" marks in that particular assignment.

Recourse exam / Make-up examination/work policy: In case if a student fails or wants to improve his/her grade in the course, recourse is permitted as per University academic regulations. No make-up work will be given for unexcused absences. The faculty needs to be informed in advance in case the student is not going to be able to submit an assignment or take any evolution component, and it is at the discretion of the faculty to sanction make up for an evaluation component.

Behavior expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic dishonesty/cheating/plagiarism: Plagiarism and dishonesty in any form in any Evaluation component will lead to appropriate disciplinary action.

List of projects

S. No	Project Title
1	Design and implementation of a runtime deadlock detection mechanism by using banker's algorithm.
2	application of multithreading
3	DIGITAL CPU SCHEDULING
4	CPU Scheduling
5	DISK SCHEDULING APPPLICATION
6	monitoring CPU utilization using shall script
7	Creating Mini Operating System
8	memory management
9	case study on CPU scheduling
10	implementation of Ionic framework
11	page fault algorithms (Web)
12	CPU scheduling (App)
13	page fault (app)
14	writing a mini shell
15	basic boot loader

Student Co-Ordinator's